

Michail Tamiolakis

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SUMMARY

Unity Engineer with 4+ years of professional experience building **simulations**, **XR experiences**, and **interactive mechanics** shipped on **Meta Quest** and **Apple Vision Pro**. Main contributor to **MAGES 5.0 (NXT)**, a Unity Asset Store SDK and Unity Verified Solution. Experienced in profiling and resolving performance bottlenecks on standalone hardware, and in building tools and asset pipelines for artists and designers. MSc in Computer Science and Engineering with published research on real-time soft-body simulation.

SKILLS

Languages: C#, C++

Unity & XR: Unity, Unity DOTS, Burst Compiler, Jobs, Editor Tooling, Unity Test Framework, OpenXR, Meta SDK, Photon, XRI, MRTK, OpenXR, Addressables

Gameplay & Interaction: Interaction Systems, Hand Interaction, Input Frameworks, Physics-Based Mechanics, XPBD, Soft-Body Simulation, Real-Time Cutting, Tearing & Piercing

Asset Pipelines: Volumetric (Tetrahedral) Model Generation, 3D Artist Guidelines & Asset Budgets, LODs, Mesh & Texture Memory Optimization

Performance & Profiling: Unity Profiler (Deep Profiling), Memory Profiler, OVR Metrics, GC Allocation Reduction, Object Pooling

Platforms & Systems: Meta Quest 2/3/Pro, Apple Vision Pro, Pico, HoloLens, PC; Parallel Programming, Networking, CI/CD, Git

EXPERIENCE

Hellenic Air Force

Nov. 2025 - Aug. 2026

Mandatory Military Service

Greece

- Administrative office role; developed **Excel** (VBA) automations for HR management tasks.

ORamaVR

Sep. 2024 - Jun. 2025

Lead SDK Developer

Heraklion, Crete

- Main contributor (by subsystems shipped and lines of code written) to **MAGES 5.0 (NXT)**, a complete rewrite of the MAGES 4.0 SDK with a new architecture and systems, published on the **Unity Asset Store (Unity Verified Solution)** and used in private custom applications for hospitals and universities and in store applications for **Meta Quest** and **Apple Vision Pro**.
- Designed and implemented a hardware-agnostic, cross-platform input framework (**OpenXR**, **Meta SDK**) and a hand interaction system with **automated pose estimation**, removing manual pose creation for about 80% of interactive items and reducing simulation development time.
- Designed and implemented a soft-body physics engine based on **Extended Position-Based Dynamics (XPBD)** in a massively parallel, data-oriented architecture built on the **Burst Compiler**, **Jobs**, and **SIMD**, with real-time cutting, tearing, and piercing of soft bodies, ropes, and rods via custom constraints. Over 150% faster than the legacy solution on **Meta Quest 2, 3, Pro** and PC, and 39% faster than SOFA, a widely used open-source simulation framework, on PC (physics step time, ms).
- Worked with designers and 3D artists: built systems that generate the volumetric (tetrahedral) models required by the soft-body physics from surface meshes, and wrote guides for artists on authoring these models in external software, including maximum tetrahedra counts.
- Profiled and optimized the physics framework and interaction system with the **Unity Profiler** (including deep profiling), **Memory Profiler**, and **OVR Metrics** on standalone headsets, checking each system against the performance and memory requirements of the Meta Quest and Apple Vision Pro stores.
- Eliminated per-frame **GC allocations** in the interaction system, which created a new event every frame while an item was hovered or grabbed, by introducing an **event pool**. Together with other fixes, this raised the framerate from below 60 FPS to above 60 FPS on standalone VR hardware.
- Reduced scene triangle counts through mesh simplification and **LODs**, and lowered texture and mesh memory usage to fit standalone VR memory limits.

- Built a **Unity Editor tool** that authenticates users (login code or **Google SSO**) against the backend licensing service and enables or restricts paid SDK features by license.
- Led a sub-team of 2-5 engineers responsible for a feature group within the SDK (technical direction, planning, estimation, code reviews, onboarding), and built internal tools and CI/CD improvements that cut release and testing cycle times by about 15%.

ORamaVR

Experienced VR Developer

Apr. 2022 - Aug. 2024
Heraklion, Crete

- Designed and implemented new SDK features and interactive mechanics for simulation-specific tasks based on application requirements, with tests (**Unity Test Framework**) and **Sphinx** documentation.

ORamaVR

Junior VR Developer

Jun. 2021 - Mar. 2022
Heraklion, Crete

- Created interactive medical simulation mechanics with the **MAGES SDK**, and performed QA testing.

PROJECTS

Lego Builder - Cross-Reality Building Game (VR & AR) | [GitHub](#)

- Developed the VR game for free-form Lego-style brick construction with unlimited pieces.
- Implemented undo/redo and action recording, serializing each build step into a portable JSON sequence.
- Set up the AR project for a companion app that replays recordings as a real-world building guide.

BoFUN - Multiplayer Board Game System for a Smart Living Room | [GitHub](#)

- Developed a multiplayer, multi-device party-game platform combining a Unity-based interactive coffee table (touch interactions for dice rolls, game board, and piece placement), a wall-projected display, and a smartphone client.
- Built a real-time socket server that synchronizes game state, scores, and events across all devices, with QR-code player onboarding.
- Added voice command recognition (SRGS grammar) and text-to-speech narration for hands-free play.

RESEARCH

Progressive Tearing and Cutting of Soft-Bodies in High-Performance Virtual Reality

ICAT-EGVE 2022 - Hiyoshi, Yokohama, Japan (2022)

- Developed an optimized algorithm for real-time cuts and consecutive tears on soft-body meshes for standalone VR.

MAGES 4.0: Accelerating the World's Transition to VR Training and Democratizing the Authoring of the Medical Metaverse

IEEE Computer Graphics and Applications (2023)

- Contributed physics systems and interactive mechanics to the MAGES SDK.

EDUCATION

University of Crete

Master of Science in Computer Science and Engineering (GPA: 9.2/10)

Sep. 2023 - Mar. 2025
Heraklion, Greece

- **MSc Thesis:** A Real-Time Physics Framework with Advanced Cutting, Tearing and Piercing Interactions for Mobile XR Applications  [Demo Video](#)
- Performance case study: benchmarked the custom framework against the **SOFA** simulation framework.

University of Crete

Bachelor of Science in Computer Science and Engineering (GPA: 7.6/10)

Sep. 2017 - Mar. 2022
Heraklion, Greece

- **BSc Thesis:** An Adaptive, Optimized Soft-Body Mesh Algorithm for Interactive Skinned Geometrical Meshes in VR

LANGUAGES

Greek (Native) English (C2, ECPE) German (B2, Goethe-Zertifikat)